

Section 1: Fundamental Principles of Laser

1. **LASER stands for:**

- A) Light Absorption by Stimulated Emission of Radiation
- **B) Light Amplification by Stimulated Emission of Radiation**
- C) Light Amplitude by Static Emission of Radiation
- D) Light Absorption by Solar Emission of Radiation

2. **The concept of stimulated emission was first proposed by:**

- A) Isaac Newton
- **B) Albert Einstein**
- C) Niels Bohr
- D) Theodore Maiman

3. **Laser light is "Monochromatic," which means it has:**

- **A) Only one color (wavelength)**
- B) Many colors
- C) No color
- D) High intensity

4. **Laser light is "Coherent," meaning the light waves are:**

- A) Moving in different directions
- **B) In phase (synchronized in time and space)**
- C) Out of phase
- D) Scattered

5. **A laser beam is highly "Directional," which means it:**

- A) Spreads out quickly
- **B) Travels a long distance with very little spreading**
- C) Moves in a circle
- D) Is absorbed by air immediately

6. **The process of supplying energy to the laser medium to excite atoms is called:**

- A) Cooling
- **B) Pumping**
- C) Radiation
- D) Ionization

7. **The state where more atoms are in the excited state than in the ground state is:**

- A) Normal Distribution

- **B) Population Inversion**
 - C) Atomic Stability
 - D) Ground Equilibrium
8. **Which component of a laser system reflects light back and forth to amplify it?**
- A) The Power Supply
 - **B) Optical Resonator (Mirrors)**
 - C) The Cooling Fan
 - D) The Laser Medium
9. **Stimulated emission produces a photon that is identical to the stimulating photon in:**
- A) Frequency
 - B) Phase
 - C) Direction
 - **D) All of the above**
10. **The first working laser (Ruby Laser) was built by:**
- A) Albert Einstein
 - **B) Theodore Maiman**
 - C) Charles Townes
 - D) Arthur Schawlow

Section 2: Types of Lasers

11. **The Ruby Laser is an example of which type of laser?**
- **A) Solid-state Laser**
 - B) Gas Laser
 - C) Liquid Laser
 - D) Semiconductor Laser
12. **The active medium in a Helium-Neon (He-Ne) laser is:**
- A) Solid
 - **B) Gas**
 - C) Liquid
 - D) Plasma
13. **Which laser is commonly used in CD/DVD players and laser pointers?**
- A) CO₂ Laser
 - B) Ruby Laser
 - **C) Semiconductor (Diode) Laser**

- D) Argon Laser
- 14. **The CO₂ laser is widely used in industry for cutting because it is:**
 - **A) High power and efficient**
 - B) Very small
 - C) Blue in color
 - D) Low energy
- 15. **Dye lasers use which type of active medium?**
 - A) Rare gases
 - B) Solid crystals
 - **C) Organic dye solutions (Liquid)**
 - D) Metal vapors
- 16. **Which gas is used in "Excimer" lasers often used in eye surgery?**
 - A) Oxygen
 - **B) Noble gases mixed with reactive gases**
 - C) Carbon dioxide
 - D) Hydrogen
- 17. **The color of the light emitted by a He-Ne laser is typically:**
 - A) Green
 - **B) Red**
 - C) Blue
 - D) Invisible (Infrared)
- 18. **Nd:YAG is a popular laser in medical and industrial fields. It is a:**
 - **A) Solid-state Laser**
 - B) Gas Laser
 - C) Semiconductor Laser
 - D) Liquid Laser
- 19. **Chemical lasers derive their energy from:**
 - A) Electricity
 - **B) Chemical reactions**
 - C) Sunlight
 - D) Nuclear heat
- 20. **Which laser type is most efficient at converting electrical energy into light?**
 - A) Ruby Laser

- B) Gas Laser
- **C) Semiconductor Laser**
- D) Dye Laser

Section 3: Uses of Laser Beams (Medicine & Industry)

21. In medicine, lasers are used for "Bloodless Surgery" because they:
 - A) Are cold
 - **B) Seal blood vessels while cutting (Cauterization)**
 - C) Do not touch the skin
 - D) Use visible light only
22. The use of laser to reshape the cornea to correct vision is called:
 - **A) LASIK**
 - B) Dialysis
 - C) Endoscopy
 - D) X-ray
23. Which laser is commonly used for tattoo removal?
 - A) CO₂ Laser
 - **B) Q-switched Nd:YAG Laser**
 - C) He-Ne Laser
 - D) Diode Laser
24. In industry, lasers are used for "LIDAR," which stands for:
 - A) Light Intense Data and Recording
 - **B) Light Detection and Ranging**
 - C) Laser Internal Data Analysis
 - D) Light Induced Digital Auto Research
25. Lasers are used in "Holography" to create:
 - A) 2D Photographs
 - **B) 3D Images**
 - C) Black and white films
 - D) Digital text
26. Which property allows lasers to guide missiles and smart bombs?
 - A) Color
 - **B) High directionality (straight line)**
 - C) Heat

- D) Brightness
- 27. **Lasers are used in fiber optic communication to carry:**
 - A) Electricity
 - **B) Data signals (Voice, Internet, Video)**
 - C) Water
 - D) Sound waves directly
- 28. **In supermarkets, low-power lasers are used in:**
 - A) Refrigerators
 - **B) Barcode Scanners**
 - C) Weighing scales
 - D) Trolleys
- 29. **Laser "Drilling" is preferred over mechanical drilling for very hard materials like:**
 - A) Wood
 - B) Plastic
 - **C) Diamonds**
 - D) Paper
- 30. **Which laser is used to treat skin conditions like birthmarks or port-wine stains?**
 - **A) Pulsed Dye Laser**
 - B) CO₂ Laser
 - C) He-Ne Laser
 - D) Ruby Laser

Section 4: Communication, Research & Safety

- 31. **What makes laser ideal for measuring the distance to the Moon?**
 - A) Its brightness
 - **B) High directionality and narrow beam**
 - C) Its red color
 - D) Its heat
- 32. **Laser printers use a laser beam to:**
 - A) Burn the paper
 - **B) Create an electrostatic image on a drum**
 - C) Melt the ink directly
 - D) Scan the document
- 33. **In nuclear fusion research, high-power lasers are used to:**

- **A) Compress and heat fuel pellets**
 - B) Cool the reactor
 - C) Record the data
 - D) Generate gravity
34. **"Optical Fiber" works on the principle of:**
- A) Refraction
 - **B) Total Internal Reflection**
 - C) Dispersion
 - D) Scattering
35. **The main hazard of direct exposure to a laser beam is damage to the:**
- A) Ears
 - B) Bones
 - **C) Eyes (Retina)**
 - D) Teeth
36. **Which class of laser is considered safe for the human eye under normal use?**
- **A) Class 1**
 - B) Class 2
 - C) Class 3
 - D) Class 4
37. **Laser "Spectroscopy" is a technique used to:**
- A) Cut metal
 - **B) Study the properties of matter (atoms and molecules)**
 - C) Print labels
 - D) Watch 3D movies
38. **The "Active Medium" of a laser determines the:**
- A) Weight of the laser
 - **B) Wavelength (color) of the light**
 - C) Cost of the laser
 - D) Shape of the laser
39. **In the garment industry, lasers are used for:**
- A) Dyeing clothes
 - **B) Precision cutting of fabric**
 - C) Ironing
 - D) Sewing buttons

40. **A laser beam can be focused to a very small spot, creating extremely high:**
- A) Volume
 - **B) Intensity (Energy density)**
 - C) Speed
 - D) Frequency

Section 5: Rapid Fire Review (41-100)

41. **Medium in He-Ne laser? (Ans: Neon and Helium gas)**
42. **Light amplification occurs in? (Ans: The optical cavity/Resonator)**
43. **Spontaneous emission occurs? (Ans: Randomly without external trigger)**
44. **Laser used in dental treatment? (Ans: Er:YAG or Diode Laser)**
45. **Property of laser light being in a single path? (Ans: Collimated)**
46. **Mirror at one end of the laser is? (Ans: Fully reflective)**
47. **Mirror at the output end is? (Ans: Partially reflective)**
48. **Laser used for measuring earth's movement? (Ans: Laser Interferometer)**
49. **Main component of Ruby laser crystal? (Ans: Aluminum Oxide with Chromium)**
50. **Laser used in CD players? (Ans: Infrared Diode Laser)**
51. **Can laser travel through a vacuum? (Ans: Yes)**
52. **Laser used to treat detached retina? (Ans: Argon Laser)**
53. **Invisible laser radiation is usually? (Ans: Infrared or Ultraviolet)**
54. **Laser used for underwater communication? (Ans: Blue-Green Laser)**
55. **Laser used in the construction industry for leveling? (Ans: Laser Level)**
56. **Does laser light obey the laws of reflection? (Ans: Yes)**
57. **Process of atoms returning to ground state? (Ans: De-excitation)**
58. **Speed of laser light? (Ans: 3×10^8 m/s)**

59. **Common gas used in high-power industrial lasers? (Ans: CO₂)**
60. **Semiconductor laser material? (Ans: Gallium Arsenide)**
61. **Laser used for welding? (Ans: Nd:YAG or Fiber Laser)**
62. **Acronym for Light Detection and Ranging? (Ans: LIDAR)**
63. **Does a laser beam require a medium like air to travel? (Ans: No)**
64. **Light produced by a bulb is? (Ans: Incoherent)**
65. **Light produced by laser is? (Ans: Coherent)**
66. **A 3D laser image is called? (Ans: Hologram)**
67. **Device used to produce laser light? (Ans: Laser Oscillator)**
68. **Optical fibers are made of? (Ans: Glass or Plastic)**
69. **Laser used for clearing blocked arteries? (Ans: Excimer Laser)**
70. **Most powerful type of laser pulse? (Ans: Femtosecond Laser)**
71. **Role of Helium in He-Ne laser? (Ans: Energy transfer to Neon)**
72. **Can laser be used for hair removal? (Ans: Yes)**
73. **Laser used in high-speed photography? (Ans: Pulsed Laser)**
74. **Is laser light safer than X-rays? (Ans: Yes, it is non-ionizing)**
75. **Laser used in photolithography? (Ans: Ultraviolet Laser)**
76. **What is 'pumping' in lasers? (Ans: Supply of energy)**
77. **Which laser has the highest heat? (Ans: CO₂)**
78. **Laser used in pointers? (Ans: Diode)**
79. **Number of colors in a laser beam? (Ans: One)**
80. **Can laser carry sound? (Ans: Yes, as digital data)**
81. **Laser used in finger-print detection? (Ans: Argon Laser)**
82. **Is laser light polarized? (Ans: Usually, yes)**
83. **Smallest laser? (Ans: Nanowire/Semiconductor laser)**
84. **Laser used in laser light shows? (Ans: Argon/Krypton)**
85. **Unit of laser power? (Ans: Watt)**
86. **Can laser cut through steel? (Ans: Yes)**
87. **Laser used in kidney stone removal? (Ans: Holmium Laser)**
88. **Most common pumping method for gas lasers? (Ans: Electrical discharge)**

89. **Most common pumping for solid-state? (Ans: Optical/Flash lamp)**
90. **Laser used in land surveying? (Ans: Diode/He-Ne)**
91. **Distance a laser can travel? (Ans: Depends on power and medium)**
92. **Are lasers used in aircraft navigation? (Ans: Yes, Ring Laser Gyros)**
93. **Does laser light spread like a torch? (Ans: No)**
94. **Metastable state? (Ans: State where atoms stay longer than usual)**
95. **Importance of Metastable state in laser? (Ans: Helps in Population Inversion)**
96. **Laser used for reading Barcodes? (Ans: He-Ne or Diode)**
97. **Can lasers be used in space? (Ans: Yes)**
98. **Discovery year of first laser? (Ans: 1960)**
99. **Is laser beam a stream of electrons? (Ans: No, it's photons)**
100. **Major advantage of laser in communication? (Ans: High bandwidth/Data capacity)**